Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Technical Report - English (Svarah) and Telugu (IndicTTS Telugu)

GitHub repository: https://github.com/karankumar211/IIITH-ASSIGNMENT

1. Implementation Approach

Two separate FastSpeech 2 models were trained, one for English and one for Telugu, starting from the public reference implementation by ming024 (github.com/ming024/FastSpeech2). Each language has its own code folder, configuration (config.yaml), checkpoints and results. Every language-specific step (alignment, preprocessing, training, inference) was run independently.

1.1 Components reused from the reference implementation

- Model: 4-layer FFT encoder and 6-layer FFT decoder (hidden size 256, 2 attention heads, conv kernel [9, 1], filter size 1024), variance adaptor with duration, pitch and energy predictors and a length regulator, mel-spectrogram decoder with postnet (about 35.3 M parameters).
- Pitch and energy are modelled at phoneme level, normalised, and quantised into 256 linear bins; the variance adaptor is trained with ground-truth durations from forced alignment.
- Loss: mel (before/after postnet) + pitch + energy + duration terms; optimiser: Adam with warm-up (4,000 steps) and step decay.
- Vocoder: pre-trained universal HiFi-GAN supplied with the reference code; training/features: 22,050 Hz, FFT 1024, hop 256, 80 mel bands, fmax 8 kHz.
- Forced alignment: Montreal Forced Aligner providing the phoneme durations required by FastSpeech 2.

1.2 Modifications common to both languages

- Compatibility fixes for current libraries: librosa keyword-only arguments, removal of deprecated numpy aliases, float32 casting of pitch/energy/mel tensors (the original code crashed with a "Double vs Float" error on current PyTorch).
- MFA-3 TextGrid handling: empty intervals are treated as silence ("sp").
- New preprocess.py per language: writes the config split into config/<lang>/*.yaml, runs MFA alignment, extracts features, and re-splits train/val exactly to the utterance lists produced by data selection (guaranteeing no overlap).
- New inference.py per language: synthesis of the test sentences, training/validation loss plots from the logs, and duration/pitch/energy plots and CSV values for each test sentence. A duration-scale option (d_control) was added (see Section 5).

1.3 English-specific modifications

- Alignment with the pretrained MFA english_us_arpa acoustic model and dictionary.
- ARPAbet phoneme symbols (original symbol set). At inference, g2p_en (part-of-speech aware) converts text to phonemes so that homographs (e.g. "conduct") and out-of-vocabulary words are handled.
- Multi-speaker model (speakers labelled by gender in Svarah).

1.4 Telugu-specific modifications

- New symbol set (text/symbols.py): the Telugu Unicode block (U+0C00-U+0C7F) as grapheme tokens, plus digits, punctuation and silence symbols. Graphemes (Unicode code points) are used as acoustic units because no Telugu phoneme dictionary or pretrained aligner was available.
- Alignment: a grapheme dictionary (one entry per word, graphemes as phones) was built from the training transcripts and an MFA acoustic model was trained on the Telugu data itself. To keep it tractable, a monophone-only configuration with 20 iterations was used.

- Text normalisation: Unicode NFC, removal of zero-width joiners, removal of non-Telugu characters; utterances containing digits were excluded from training.
- The dataset contains two speakers, so the Telugu model is multi-speaker (speaker id 0 used at inference).

2. Datasets, Subsampling and Preprocessing

2.1 Selection strategy

- English: ai4bharat/Svarah (Hugging Face), 6,656 utterances of Indian-accented English with read and spontaneous speech. Telugu: SPRINGLab/IndicTTS_Telugu (Hugging Face), derived from the IIT Madras IndicTTS database.
- Utterances were shuffled with a fixed seed (42) and added one by one until about 5 h (train) and 1 h (validation) of audio were reached, so the selection is a random utterance-level subsample.
- Filters: duration 1-11 s (stable alignment, fits the 1200-frame limit); text cleaning (lower-casing, bracketed tags and non-letters removed for English; Telugu script only for Telugu); utterances containing digits dropped because they cannot be aligned reliably.
- Each utterance is assigned to either train or validation, never both; this is asserted in the code. The exact utterance lists are stored in data_split/<Language>_split.json.

2.2 Resulting data

Language	Dataset	Train utts	Val utts	Train h (selected)	Val h (selected)	Train h after alignment	Val h after alignment	Speakers
English	Svarah	3,380	691	4.71	0.97	3.62	0.75	2 (female, male)
Telugu	IndicTTS Telugu	2,613	525	5.00	1.00	3.51	0.69	2

"Selected" is the raw audio chosen by the sampling procedure (about 5 h / 1 h as required). "After alignment" is the speech that remains after MFA trimming of leading/trailing silence and removal of a few utterances that failed alignment (English: 3,370 train / 686 val; Telugu: 2,601 train / 520 val). The effective amount of speech used for training is therefore lower than the nominal 5 h / 1 h.

2.3 Preprocessing steps

- Resampling to 22,050 Hz, peak normalisation, 16-bit WAV; one .lab transcript per utterance.
- MFA alignment -> TextGrids -> per-phoneme durations; mel-spectrogram, pitch (PyWorld) and energy extraction at phoneme level; global pitch/energy normalisation statistics (stats.json).

3. Training Configuration

Setting	English	Telugu
Total steps	40,000	40,000
Batch size / grad. accumulation	16 / 1	16 / 1
Optimiser	Adam (0.9, 0.98), eps 1e-9	same
LR schedule	warm-up 4,000 steps; decay x0.3 at 70% and 85% of total steps	same
Checkpoints / validation	every 10,000 steps / every 1,000 steps	same
Alignment / units	MFA pretrained, ARPAbet phonemes	MFA trained on data, graphemes
Speakers	2 (multi-speaker)	2 (multi-speaker)

Setting	English	Telugu
Hardware	Kaggle NVIDIA T4 GPU	Kaggle NVIDIA T4 GPU

A 300-step smoke test was run first to validate the pipeline; its checkpoints were discarded before the full run. The final checkpoints (step 40,000) are used for all results.

4. Loss Curve Analysis

Model	Train total loss (step ~100)	Train total loss (step 40,000)	Val total loss (step ~100)	Val total loss (step 40,000)	Lowest val loss
English	9.69	0.96	8.52	4.12	3.91 (step 1,000)
Telugu	9.92	0.82	9.01	4.05	3.82 (step 2,000)

- Convergence: training loss falls quickly in the first thousands of steps and then keeps decreasing slowly; the final train mel loss is about 0.46 (English) and 0.39 (Telugu).
- Overfitting: validation total loss stops improving very early. After step 5,000 it stays flat at about 4.05-4.13 for both languages (English 4.07 at 10k, 4.13 at 20k, 4.12 at 40k; Telugu 4.06 at 5k, 4.05 at 20k, 4.05 at 40k), while training loss keeps dropping. The validation mel loss (about 1.04 English, 0.98 Telugu) is more than twice the training mel loss. The training/validation gap is large, i.e. the models memorise the roughly 3.5 h of speech used for training; no instability or divergence was observed.
- Consequence for checkpoint choice: since validation loss is flat after 5,000 steps, later checkpoints are not clearly better on unseen data, and the intermediate checkpoints that would have had the lowest validation loss (steps 1,000-2,000) were not saved. The final step-40,000 checkpoint was used.

The available training summaries report total loss and mel-loss values, but do not provide a complete, consistently recorded component-wise train/validation series for mel, pitch, energy and duration. Therefore, the total-loss gap cannot be attributed quantitatively to one specific variance-adaptor term. The reported evidence supports the conclusion that training loss continued to decrease while validation total loss plateaued; identifying the component responsible would require the corresponding component-wise log entries.

Loss-curve figures: include the English and Telugu training_loss.png and validation_loss.png files under Results/<Language>/. The curves are based on the recorded log entries, not on every optimizer update.

5. Generated Speech - Observations per Test Sentence

The final recovered project contains five synthesized test audios per language. Important evaluation note: the final audio set does not match the assignment's prescribed E1-E4 and T1-T4 sentences. The English set used was: (1) “Hello, how are you today?”, (2) “The researchers will conduct an experiment.”, (3) “Students learn new things every day.”, (4) “And they went off into the space when earth was burning up or being destroyed.”, and (5) “Karnataka's expansive list of handmade artisanal products are impressive.” The Telugu set used was: (1) “నమస్కారం, మీరు ఎలా ఉన్నారు?”, (2) “నేను బాగున్నాను”, (3) “ఈ రోజు చాలా మంచి రోజు.”, (4) “విద్య మన జీవితంలో చాలా ముఖ్యం.”, and (5) “మనమందరం కలిసి పని చేయాలి.” This deviation must be disclosed rather than presenting these samples as the assignment's exact test sentences.

- Duration scaling: the default duration scale produced speech that was noticeably too fast, particularly for longer inputs. The duration control was therefore increased to 1.5 (d_control = 1.5) for the final generation run. This is an empirical inference adjustment; it does not change the trained checkpoint.

- The final test set listed above is the set present in the recovered project outputs. It should not be described as the assignment's exact sentence set. For strict compliance, the prescribed assignment sentences must be synthesized and evaluated separately.

ID	Sentence	Expected difficulty	Observation (pronunciation, speed, clarity)
EN-1	Hello, how are you today?	Short greeting	Reported as generally understandable; no separate sentence-level defect log was retained.
EN-2	The researchers will conduct an experiment.	Sentence with the word “conduct”	English output was generally understandable; occasional pronunciation/clarity variation was reported across the set.
EN-3	Students learn new things every day.	Short declarative sentence	No sentence-specific listening note was retained; overall English samples were generally understandable.
EN-4	And they went off into the space when earth was burning up or being destroyed.	Long sentence from Svarah validation text	Longer input; duration scaling was used to reduce overly fast speech. No separate word-level error note was retained.
EN-5	Karnataka's expansive list of handmade artisanal products are impressive.	Longer sentence with less common vocabulary	Longer input and less common words; occasional English pronunciation/clarity variation was reported at set level.
TE-1	నమస్కారం, మీరు ఎలా ఉన్నారు?	Greeting / question	Reported as clear during listening review.
TE-2	నేను బాగున్నాను	Short declarative sentence	Reported as clear during listening review.
TE-3	ఈ రోజు చాలా మంచి రోజు.	Short declarative sentence	Reported as clear during listening review.
TE-4	విద్య మన జీవితంలో చాలా ముఖ్యం.	Declarative sentence	Reported as clear during listening review.
TE-5	మనమందరం కలిసి పని చేయాలి.	Declarative sentence	Reported as clear during listening review.

Listening observations: the five Telugu samples were reported during the project review as clear and understandable. The English samples were generally understandable, although occasional pronunciation and clarity variation was noticed. No reliable sentence-by-sentence listening log was retained, so it would be misleading to label a particular sentence as mumbled, skipped or mispronounced. These observations apply to the final project test set listed above, not to the assignment's ungenerated prescribed sentences.

6. Variance Adaptor Analysis (Duration, Pitch, Energy)

For each test sentence, the inference run extracted predicted duration, pitch and energy sequences from the variance adaptor and stored the predictions in language-level JSON analysis files. The project also contains duration, pitch and energy plots. In the recovered outputs, pitch and energy are model-output/normalized values; the original inference normalization statistics needed to convert them reliably to physical F0 in Hz and absolute energy units were not available. Consequently, these plots must not be labelled as calibrated Hz or physical energy measurements.

Interpretation of the available variance-adaptor results: duration predictions determine how many acoustic frames each input unit contributes through the length regulator. Longer or more complex inputs are expected to receive more total predicted duration, while short function words generally receive less. Pitch and energy predictions represent the model's learned prosodic contours; local rises and falls indicate relative variation in the model output, not independently measured fundamental frequency or loudness. Because the recovered report package does not contain the per-token CSV tables or the complete per-sentence prediction arrays in a form that can be checked here, exact sentence-level totals, token-duration averages, pitch ranges and energy peaks cannot be stated responsibly. The plots should therefore be discussed qualitatively, and the normalized-unit limitation should accompany every pitch/energy figure.

7. Comparison of English and Telugu Models, Limitations and Possible Improvements

- Data: only about 3.5-3.6 h of effective speech per language after alignment trimming; both models overfit (validation loss flat after about 5k steps).
- English (Svarah) is spontaneous, accented and noisy speech recorded in varied conditions, which is much harder for TTS than the studio-quality IndicTTS Telugu data.
- Alignment quality: English used a pretrained aligner; Telugu used a monophone grapheme aligner trained on only about 6 h, so Telugu durations are less accurate. Graphemes are an imperfect acoustic unit for Telugu (e.g. conjuncts and inherent vowels), which limits pronunciation accuracy.
- Both models were trained only on utterances of 1-11 s; long test sentences (E1, E4, T1, T4) are outside the training length range and required a duration scale factor.
- Digits, times and unseen words (E4, T4, "antidisestablishmentarianism", "andaroo") are not covered by the training text and are expected failure cases.

Observed failure analysis is limited by the final test-set mismatch described in Section 5. The recovered listening notes support generally clear Telugu output and generally understandable English output with occasional pronunciation/clarity variation. They do not establish whether the assignment's rare word, homograph, malformed “rhyth m”, time expression, digits or Telugu conjunct cases were synthesized correctly, because those prescribed test sentences are not present in the final recovered test set. These cases remain unverified rather than confirmed successes or failures.

Possible improvements: more data (the full Svarah and IndicTTS sets, or other Telugu corpora), data augmentation, early stopping or lower-capacity/regularised models, a Telugu G2P/phoneme lexicon instead of graphemes, a better-trained Telugu aligner (full MFA pipeline), text normalisation for numbers and times, and a vocoder fine-tuned on the target speakers.

8. Reproducibility

- Software: Python 3, PyTorch, librosa, PyWorld, g2p_en, tgt, pyyaml, datasets, matplotlib, Montreal Forced Aligner 3.4.x (conda-forge), reference FastSpeech2 code (ming024) with the patches described above. See requirements.txt.
- Hardware used: Kaggle notebook with an NVIDIA T4 GPU.

Commands (run inside English/ and Telugu/ respectively):

```
python preprocess.py                        # writes config/<lang>/*.yaml, runs MFA alignment, extracts features, splits train/val

python train.py -p config/<lang>/preprocess.yaml -m config/<lang>/model.yaml -t config/<lang>/train.yaml

D_CONTROL=1.5 python inference.py --out ../Results/<lang>   # test audio, loss plots, variance-adaptor plots
```

HiFi-GAN: unzip hifigan/generator_universal.pth.tar.zip inside hifigan/. Checkpoints (step 40,000) are in <lang>/checkpoints/ (Git LFS or external link, see README). Split files: <lang>/data_split/<lang>_split.json.

Submission compliance note: Before final submission, regenerate the ten audios using the exact assignment-provided English and Telugu sentences, then replace the current test-audio table and add per-sentence duration, pitch and energy

plots. The current report transparently documents the outputs that were actually recovered; it does not claim that the prescribed test-sentence requirement has been met.

9. Conclusion

Two separate FastSpeech 2 models for English and Telugu were trained to 40,000 steps and used to synthesize five test sentences per language. Training loss decreased substantially, while validation total loss plateaued early, indicating a substantial generalization gap with the limited aligned speech. Duration scaling improved the speaking rate for longer inputs. The recovered project includes language-specific code, configurations, checkpoints, logs, test audio and variance-adaptor analyses. However, the final test sentences differ from those prescribed in the assignment, and the available pitch/energy plots are in normalized model-output units. These limitations are stated explicitly so that the results are reproducible and are not overstated.